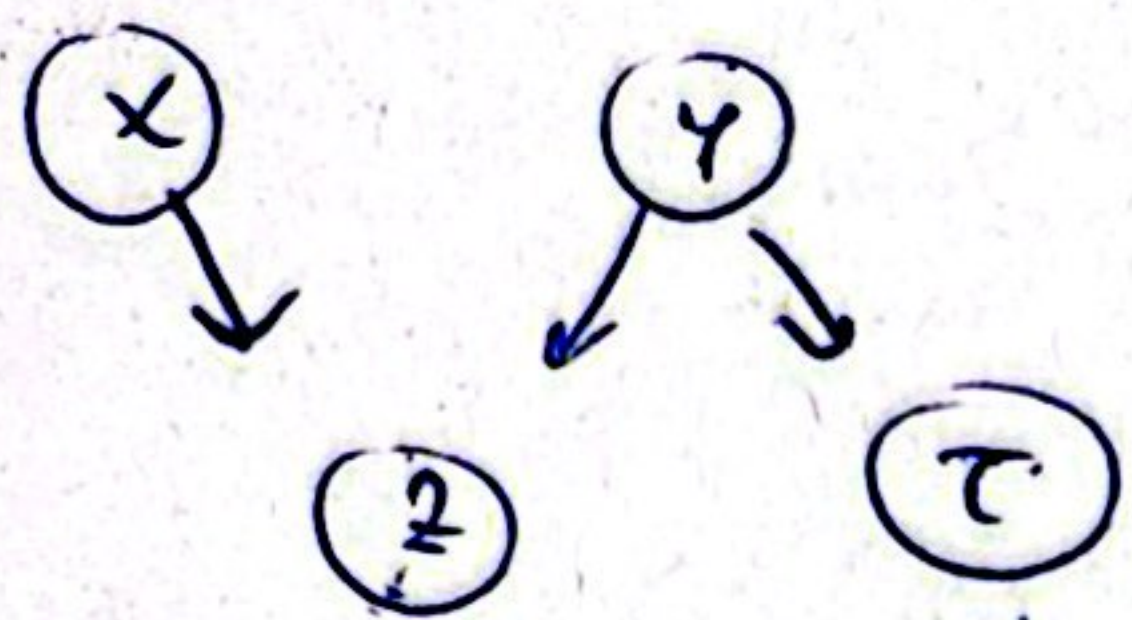




a) Paramètres no redondants.

$$\begin{aligned}\theta_{11} &= P(X=1) \\ \theta_{21} &= P(Y=1) \\ \theta_{31} &= P(Z=1/X=1, Y=1) \\ \theta_{32} &= P(Z=1/X=1, Y=0) \\ \theta_{33} &= P(Z=1/X=0, Y=1) \\ \theta_{34} &= P(Z=1/X=0, Y=0) \\ \theta_{41} &= P(T=1/Y=1) \\ \theta_{42} &= P(T=1/Y=0)\end{aligned}$$

total: 8 paramètres.

$$\begin{aligned}X &\rightarrow 1 \\ Y &\rightarrow 1 \\ Z &\rightarrow 2 \times 2 = 4 \\ T &\rightarrow 2\end{aligned}$$

2

b) $\hat{\theta}_{11} = 0.6$

$\hat{\theta}_{21} = 0.7$

$\hat{\theta}_{31} = \frac{2}{5}, \hat{\theta}_{32} = \frac{1}{1} = 1, \hat{\theta}_{33} = \frac{1}{2}, \hat{\theta}_{34} = \frac{1}{2}$ 2

$\hat{\theta}_{41} = \frac{3}{7}, \hat{\theta}_{42} = \frac{2}{3}$

c) $P(X=1, Y=1, Z=0, T=1) = P(T=1/Y=1) \cdot P(Z=0/X=1, Y=1) \cdot P(X=1) \cdot P(Y=1)$

regle de la cascade

$= \theta_{41} \cdot (1 - \theta_{31}) \cdot \theta_{11} \cdot \theta_{21}$

L'estimation est: $\hat{\theta}_{41} \cdot (1 - \hat{\theta}_{31}) \cdot \hat{\theta}_{11} \cdot \hat{\theta}_{21} = \frac{3}{7} \times \frac{2}{5} \times 0.6 \times 0.7 = \underline{0.108}$ 2

d) $L(\pi) = \prod_{i=1}^n P(\text{cas } i \text{ est vu}) =$

D = base de données

$$\begin{aligned}&= \theta_{41} \cdot (1 - \theta_{31}) \cdot \theta_{11} \cdot \theta_{21} \times ((1 - \theta_{41}) \cdot \theta_{31} \cdot \theta_{11} \cdot \theta_{21})^2 \times \\&\times (1 - \theta_{41}) \cdot (1 - \theta_{31}) \cdot \theta_{11} \cdot \theta_{21} \times \theta_{42} \cdot \theta_{32} \cdot \theta_{11} \cdot (1 - \theta_{21}) \times \\&\times \theta_{41} \cdot (1 - \theta_{31}) \cdot \theta_{11} \cdot \theta_{21} \times \theta_{42} \cdot \theta_{34} \cdot (1 - \theta_{41}) \cdot (1 - \theta_{21}) \times \\&\times (1 - \theta_{41}) \cdot \theta_{33} \cdot (1 - \theta_{11}) \cdot \theta_{21} \times (1 - \theta_{42}) \cdot (1 - \theta_{34}) \cdot (1 - \theta_{41}) \cdot (1 - \theta_{21}) \times \\&\times \theta_{41} \cdot \theta_{33} \cdot (1 - \theta_{41}) \cdot \theta_{21}\end{aligned}$$

L'estimateur de la p. de vraisemblance:

$$\hat{L}_D(\pi) = \left(\frac{3}{7} \times \frac{3}{5} \times 0.6 \times 0.7 \right) \times \left(\frac{4}{7} \times \frac{2}{5} \times 0.6 \times 0.7 \right) \times$$

$$\times \left(\frac{4}{7} \times \frac{3}{5} \times 0.6 \times 0.7 \right) \times \left(\frac{2}{3} \times 1 \times 0.6 \times 0.3 \right) \times$$

$$\times \left(\frac{3}{7} \times \frac{3}{5} \times 0.6 \times 0.7 \right) \times \left(\frac{2}{3} \times \frac{1}{2} \times 0.4 \times 0.3 \right) \times$$

$$\times \left(\frac{4}{7} \times \frac{1}{2} \times 0.4 \times 0.7 \right) \times \left(\frac{1}{3} \times \frac{1}{2} \times 0.4 \times 0.3 \right) \times$$

$$\times \left(\frac{3}{7} \times \frac{1}{2} \times 0.4 \times 0.7 \right) =$$

$$= 0.108 \times 0.096^2 \times 0.166 \times 0.12 \times 0.108 \times 0.04 \times$$

$$\times 0.08 \times 0.02 \times 0.06 =$$

$$= 7.132880359 \times 10^{-12}$$

2

$$\hat{AIC} = \ln(\hat{L}_D(\pi)) - \frac{d}{2} \ln \left(\frac{1}{10} \right) = \ln(7.132880359 \times 10^{-12}) - \frac{8}{2} \times \ln(10)$$

$d=8$

$$= -25.66630599 - 4 \times \ln(10) = -34.87664636$$

1

$$\hat{AIC} = \ln(\hat{L}_D(\pi)) - d = \ln(7.132880359 \times 10^{-12}) - 8 =$$

$$= -33.66630599$$

1

2) $D = \text{data set}$

$T = \text{DAG}$

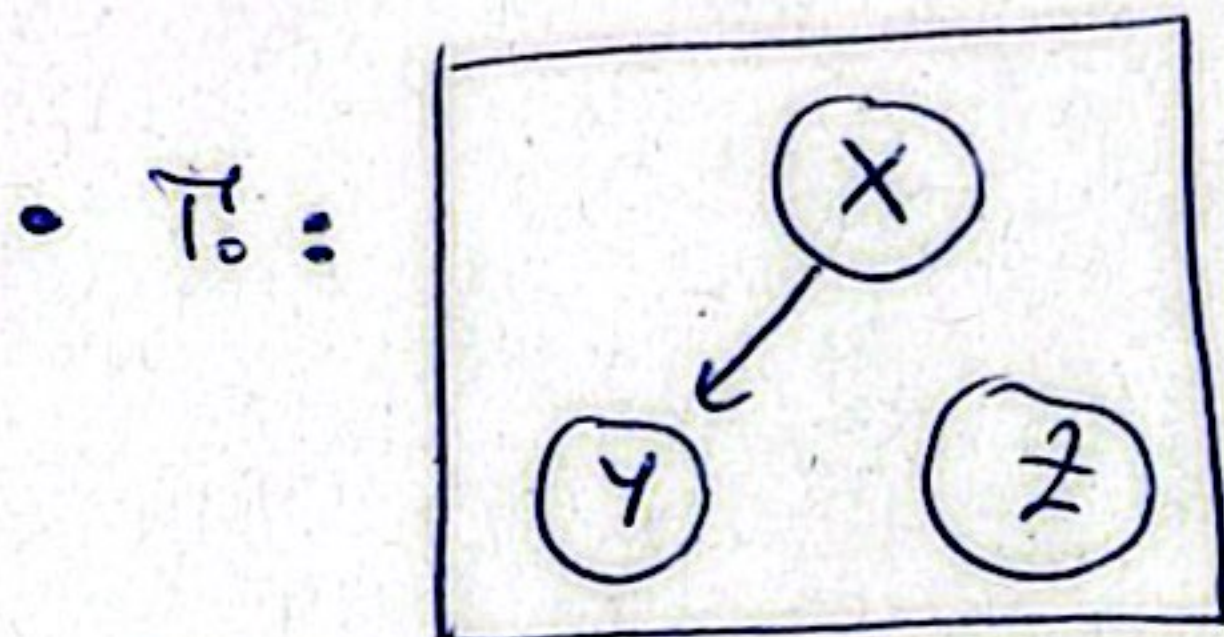
$\theta^T = \text{parameters}$

$$n = \#D = 10.$$

Kisah. Tawak - Boc D

$$L^D(\theta^T) = P(D/\theta^T) = \prod_{i=1}^n P(x_i/\theta^T) = \prod_{i=1}^n P(X=x_i, Y=y_i, Z=z_i/\theta^T)$$

$x_i = (x_i, y_i, z_i)$



$$\theta^{T_0} = (\theta_{11}, \theta_{21}, \theta_{22}, \theta_{31}) = P(X=1) P(Y=1/X=1) P(Y=1/X=0) P(Z=1)$$

$$L^D(\theta^{T_0}) = \prod_{i=1}^n P(y_i/x_i/\theta^{T_0}) \cdot P(x_i/\theta^{T_0}) \cdot P(z_i/x_i/\theta^{T_0})$$

$$= (\theta_{21}^4 \cdot \theta_{11} \cdot \theta_{31}) \cdot (1-\theta_{21}) \cdot \theta_{11} \cdot \theta_{31} \cdot (\theta_{22} \cdot (1-\theta_{11}) \cdot \theta_{31}) \cdot (1-\theta_{22}) \cdot (1-\theta_{11}) \cdot (1-\theta_{31})$$

$$= \theta_{11}^5 \cdot (1-\theta_{11})^5 \cdot \theta_{21}^4 \cdot (1-\theta_{21}) \cdot \theta_{22}^2 \cdot (1-\theta_{22})^3 \cdot \theta_{31}^6 \cdot (1-\theta_{31})^4$$

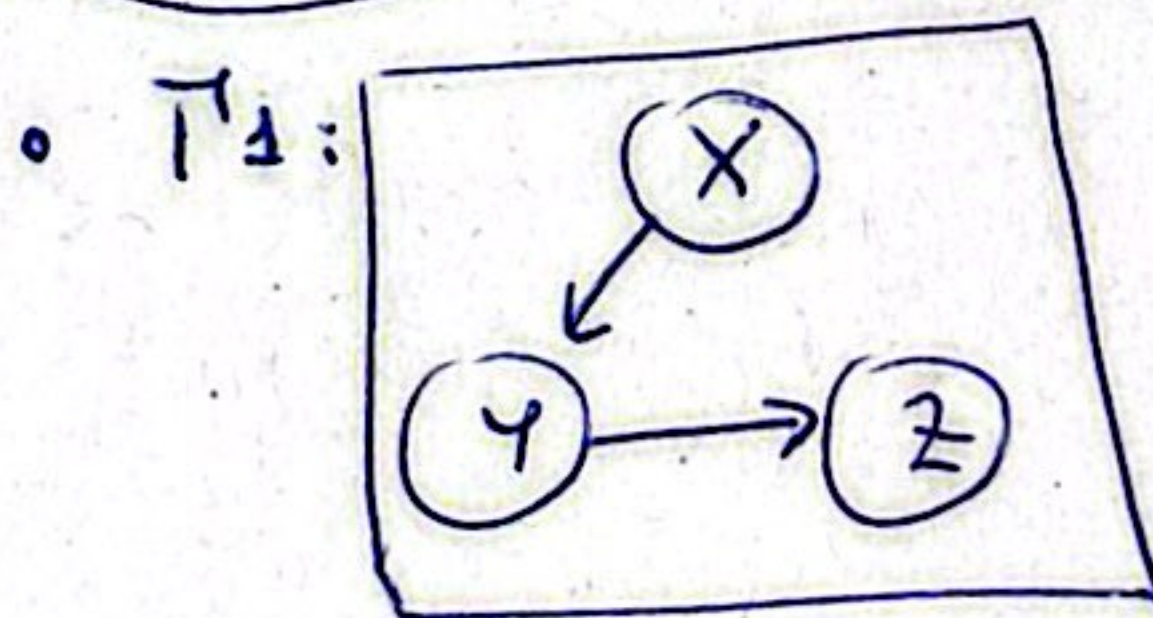
$$\hat{\theta}_{MLE}^{T_0} = (0.5, \frac{4}{5}=0.8, \frac{2}{5}=0.4, 0.6)$$

$$L^D(\hat{\theta}_{MLE}^{T_0}) = (0.5)^5 \times (0.5)^5 \times (0.8)^4 \times (0.2) \times (0.4)^2 \times (0.6)^3 \times (0.6)^6 \times (0.4)^4$$

$$= (0.5)^{10} \times 0.8^4 \times 0.2 \times (0.4)^6 \times (0.6)^9 = 3.302259425 \times 10^{-9}$$

$$BIC(T_0, D) = \ln(L^D(\hat{\theta}_{MLE}^{T_0})) - \frac{1}{2} \frac{\ln(n)}{\ln(10)}$$

$$= -19.52865893 - \frac{4}{2} \frac{\ln(10)}{\ln(10)} = -24.13382911$$



$$\theta^{T_1} = (\theta_{11}, \theta_{21}, \theta_{22}, \theta_{31}, \theta_{32}) = P(X=1) P(Y=1/X=1) P(Y=1/X=0) P(Z=1/Y=1) P(Z=1/Y=0)$$

$$L^D(\theta^{T_1}) = \prod_{i=1}^n P(z_i/y_i/\theta^{T_1}) \cdot P(y_i/x_i/\theta^{T_1}) \cdot P(x_i/\theta^{T_1})$$

$$= (\theta_{31} \cdot \theta_{21} \cdot \theta_{11})^4 \cdot (\theta_{32} \cdot (1-\theta_{21}) \cdot \theta_{11}) \cdot (\theta_{31} \cdot \theta_{22} \cdot (1-\theta_{11})) \cdot ((1-\theta_{32}) \cdot (1-\theta_{22}) \cdot (1-\theta_{11}))^3$$

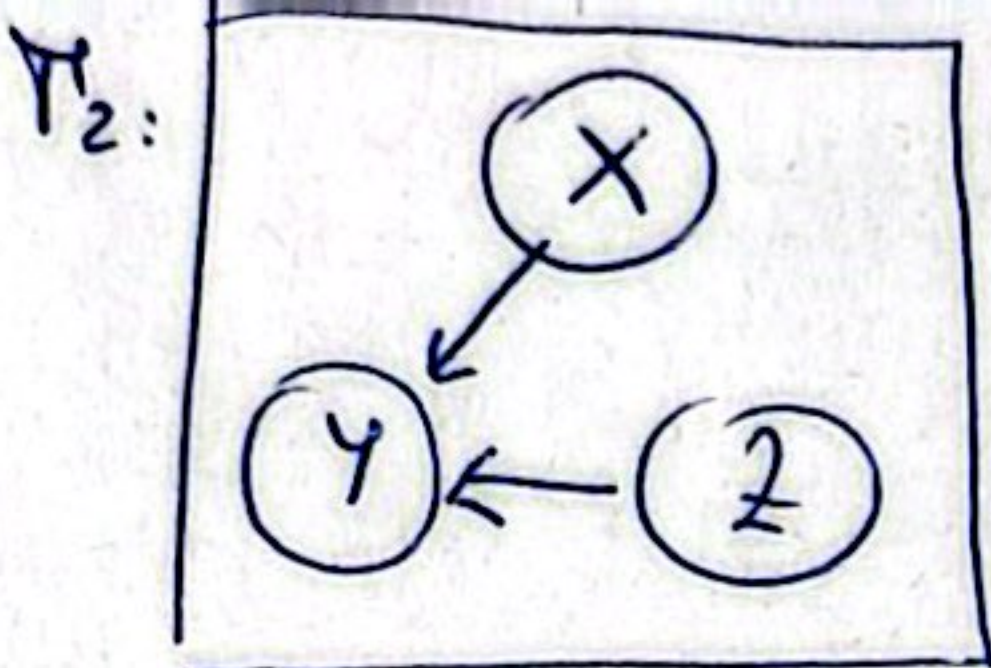
$$\cdot ((1-\theta_{31}) \cdot \theta_{22} \cdot (1-\theta_{11})) = \theta_{31}^5 \cdot (1-\theta_{31}) \cdot \theta_{32} \cdot (1-\theta_{32})^3 \cdot \theta_{21}^4 \cdot (1-\theta_{21}) \cdot \theta_{22}^2 \cdot (1-\theta_{22})^3$$

$$\cdot \theta_{11}^5 \cdot (1-\theta_{11})^5$$

$$\hat{\theta}_{MLE}^{T_1} = (0.5, 0.8, 0.4, \frac{5}{6}, \frac{1}{4})$$

$$L^D(\hat{\theta}_{MLE}^{T_1}) = (\frac{5}{6})^5 \times \frac{1}{6} \times 0.25 \times (0.75)^3 \times 0.8^4 \times 0.2 \times 0.4^2 \times 0.6^3 \times 0.5^5 \times 0.5^5 = 1.953125 \times 10^{-8}$$

$$BIC(T_1, D) = \ln(L^D(\hat{\theta}_{MLE}^{T_1})) - \frac{1}{2} \frac{\ln(n)}{\ln(10)} = -23.50771282$$



$\theta^{\pi_2} = (\theta_{11}, \theta_{21}, \theta_{22}, \theta_{31}, \theta_{32})$

$P(X=1) \quad P(Y=1/X=1, Z=1) \quad P(Y=1/X=1, Z=0) \quad P(Y=0/X=0, Z=1)$

$\theta_{24}, \theta_{31}) \quad P(Y=1/X=0, Z=0) \quad P(Z=1)$

$\boxed{H=6}$

$$L(\theta^{\pi_2}) = \prod_{i=1}^n P(Y=y_i/X=x_i, Z=z_i) \cdot P(X=x_i) \cdot P(Z=z_i) =$$

$$= (\theta_{21} \cdot \theta_{11} \cdot \theta_{31})^4 \cdot ((1-\theta_{21}) \cdot \theta_{11} \cdot \theta_{31}) \cdot (\theta_{23} \cdot (1-\theta_{11}) \cdot \theta_{31}) \cdot ((1-\theta_{24}) \cdot (1-\theta_{11}) \cdot (1-\theta_{31}))^3$$

$$\cdot (\theta_{24} \cdot (1-\theta_{11}) \cdot (1-\theta_{31})) =$$

$$= \theta_{11}^5 \cdot (1-\theta_{11})^5 \cdot \theta_{21}^4 \cdot (1-\theta_{21}) \cdot \theta_{23} \cdot \theta_{24} \cdot (1-\theta_{24})^3 \cdot \theta_{31}^6 \cdot (1-\theta_{31})^4$$

$$\hat{\theta}_{MLE}^{\pi_2} = (0.5, \frac{4}{5}, \frac{0}{0}, \frac{1}{1}, \frac{1}{1}, 0.6)$$

$\swarrow$
0.8

$\searrow$
0.25

en $\hat{\theta}_{22}$ no nos
buen falta!!!

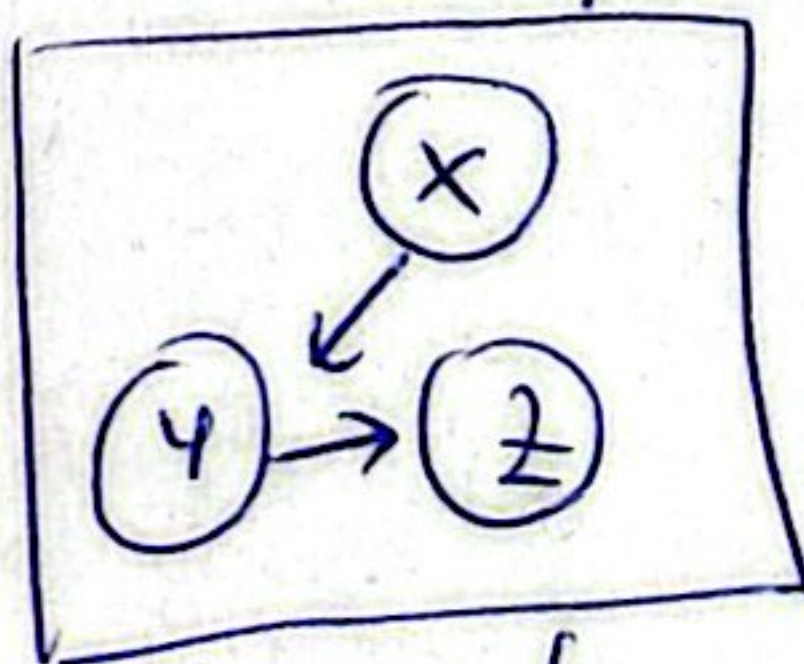
$$L(\hat{\theta}_{MLE}^{\pi_2}) = 0.5^5 \times (0.5)^5 \times (0.8)^4 \times (0.2) \times 1 \times 0.25 \times (0.75)^3 \times (0.6)^6 \times (0.4)^4 =$$

$$= (0.5)^{10} \times (0.8)^4 \times (0.2) \times (0.25) \times (0.75)^3 \times (0.6)^6 \times (0.4)^4 = 4.0077696 \times 10^{-8}$$

$$BIC(\pi_2, D) = \underbrace{\ln(L(\hat{\theta}_{MLE}^{\pi_2}))}_{-48.4129417} - \underbrace{\left(\frac{1}{2}\right) \ln(H)}_{\frac{3}{2}} = -25.32069645 \quad \underline{\underline{3}}$$

Bayes log-lik $\rightarrow \pi_2$

Bayes BIC $\rightarrow \pi_2 \Rightarrow$ la red (probable) tiene por estructura π_0 :



y las estimaciones MLE de los parámetros son:

$$\left. \begin{aligned} \theta_{11} &= P(X=1), & \hat{\theta}_{11,MLE} &= 0.5 \\ \theta_{21} &= P(Y=1/X=1), & \hat{\theta}_{21,MLE} &= 0.8 \\ \theta_{22} &= P(Y=1/X=0), & \hat{\theta}_{22,MLE} &= 0.4 \\ \theta_{31} &= P(Z=1/Y=1), & \hat{\theta}_{31,MLE} &= 5/6 \\ \theta_{32} &= P(Z=1/Y=0), & \hat{\theta}_{32,MLE} &= 1/4 \end{aligned} \right\}$$

1

$$\text{BIC}(\pi, D) = \sum_{i=1}^n g(x_i, \text{PAX}_i, D)$$

$$n = n^{\text{varia ble}} (= 3)$$

$n = n^{\circ}$ variable ($= 3$)
 $r_i = n^{\circ}$ de estados de la variable i ($= 2$)

$r_i = n_i$ de estado de la variable $i (= 1, 2, \dots, k)$
 $g_i = n_i$ de posible configuraciones de los padres de X_i en T .
 v_i de acuerdo valor k y P_i

$r_i = n_i$ de estados de la variable i ($i = 1, \dots, m$)
 $g_i = n_i$ de posibles configuraciones de los padres de X_i en T .
 $N_{ijk} = n_i$ de casos en T con X_i tomando valor k y PA_{X_i} en la configuración j -ésima.
 n_i de casos en T con PA_{X_i} en la configuración j -ésima.

$$N_{ij} = \sum_{k=1}^{n_i} N_{ijk} \quad (= n_i \text{ cases in } D \text{ on } PAx_i \text{ on } PAx_j)$$

$$T_1: \text{Bit}(T_1, D) = \underbrace{g(\langle x, \emptyset \rangle, D) + g(\langle y, x \rangle, D)}_{\text{équal que en } T_0} + \boxed{g(\langle z, y \rangle, D)}$$

$$\text{Bit}(T_{1 \times D}) = -23.50771282$$

$$\Delta BIC = +0.6261162919$$

$$(\pi_1 - \pi_0) (= g(\langle z, \gamma \rangle, D) - g(\langle z, \gamma \rangle, D)).$$

$$\rightarrow = 4 \times \ln\left(\frac{4}{5}\right) + 1 \times \ln\left(\frac{1}{5}\right) + \underbrace{1 \times \ln\left(\frac{1}{2}\right)}_0 + 1 \times \ln\left(\frac{1}{4}\right) + 3 \times \ln\left(\frac{3}{4}\right) - \left(\frac{2-1}{2}\right) \times 2^2 \times \ln(10) =$$

$$= \boxed{-9.356522882}$$

$$\text{Bit}(T_2, D) = -25.32069645$$

$$\Delta BIC = -1.186867336$$

$$(T_2 - T_0) = g(\langle Y, (X, Z) \rangle, D) - g(\langle Y, X \rangle, D)$$